

Colin Shea O'Brien

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Senior Backend Software Engineer with 8+ years designing and scaling cloud-native, service-oriented platforms. Deep experience building resilient microservices in Go, Node.js, and C#, developing critical billing, authentication, and API platforms, and deploying containerized applications to Kubernetes and AWS ECS on GCP and AWS using Terraform. Passionate about designing robust backend systems.

Work Experience

SENIOR SOFTWARE ENGINEER II – SchoolAI

Mar - Aug 2026

- Architected and built a knowledge graph powering personalized context for internal AI systems.
- Automated creation of ephemeral Kubernetes environments, reducing feedback cycles and improving confidence in production releases for 40 engineers.
- Created automated cost-monitoring tooling that reduced Datadog spending by 50% through proactive anomaly detection.

SENIOR SOFTWARE ENGINEER – AgentSync

July 2024 - Dec 2025

- Launched API webhook system to eliminate customer polling of critical systems and bring in major customers, including Berkshire Hathaway and AmeriLife Insurance
- Migrated complex insurance eligibility logic out of Salesforce to brand new microservice
- Eliminated days of recurring development time by publishing Docker image to handle dependency setup for 60 engineers and 15+ microservices
- Mentored 3 junior engineers in Go microservice development and Terraform infrastructure management

SOFTWARE ENGINEER II | PROJECT LEAD – Hudl

Apr 2023 - July 2024

- Improved performance of searches by up to 800% by leading migration from MongoDB to OpenSearch
- Reduced latency on critical feature by 1000% by catching a bug that made dozens of unnecessary DB calls
- Planned zero-downtime technical process to break down a single monolith service into 6 microservices

SOFTWARE ENGINEER I/II – Route App

Jul 2020 - Dec 2022

- Led the design, implementation, and migration to an improved serverless billing system leveraging AWS Lambda, RDS, SQS, and SNS to provide scalability at negligible cost
- Developed ingestion platform to securely process 20 million customer emails per day without any PII leaks, while also building a highly scalable and performant data publishing system
- Guided design and implementation of 3 unique AuthN/AuthZ platforms to allow users to securely access sensitive internal systems and data via OAuth

ASSOCIATE SOFTWARE ENGINEER – Imagine Learning, Inc.

Jan 2019 - Apr 2020

WEB DEVELOPER – Brigham Young University

Aug 2017 - Jan 2019

Technical Skills

- **Expert:** Go (GoLang), Terraform, AWS, Docker, GitLab CI, Kafka, Git, Redis, REST
- **Proficient:** Kubernetes, NodeJS, TypeScript C#, MySQL, PostgreSQL, DynamoDB, Linux systems
- **Experienced:** ReactJS, Java, C++, C, Ruby on Rails, Rust

Education

Brigham Young University

B.S. Computer Engineering, Minor in Computer Science